Să se demonstreze relația de recurență și ortogonalitatea pentru polinoamele Cebîșev de speța I, $T_n(x) = \cos(n \arccos x)$

Rezolvare

$$T_0(x) = \cos(0 \cdot \arccos x) = \cos 0 = 1$$

$$T_1(x) = \cos(\arccos x) = x$$

Pentru $n \geq 1$ introducem substituția $\theta = \arccos x$

$$\theta \in [0, \pi] \Rightarrow T_n(x) = T_n(\theta) = \cos(n\theta)$$

$$\left.\begin{array}{l} T_{n+1}(\theta) = \cos((n+1)\theta) = \cos\theta \cdot \cos n\theta - \sin\theta \cdot \sin n\theta \\ T_{n-1}(\theta) = \cos((n-1)\theta) = \cos\theta \cdot \cos n\theta + \sin\theta \cdot \sin n\theta \end{array}\right\} \Rightarrow$$

$$\left.\begin{array}{l} \Rightarrow T_{n+1}(\theta) + T_{n-1}(\theta) = 2\cos\theta \cdot \cos n\theta \\ x = \cos\theta, \quad T_n(\theta) = \cos n\theta \end{array}\right\} \Rightarrow$$

$$\Rightarrow T_{n+1}(\theta) = 2\cos\theta \, T_n(\theta) - T_{n-1}(\theta)$$

$$\boxed{\begin{array}{l} T_{n+1}(x) = 2x\,T_n(x) - T_{n-1}(x) \\ T_0(x) = 1, \quad T_1(x) = x \end{array}}$$

$$T_2(x) = 2x T_1(x) - T_0(x) = 2x^2 - 1$$

$$T_3(x) = 2x T_2(x) - T_1(x) = 4x^3 - 3x$$

$$T_4(x) = 2x T_3(x) - T_2(x) = 8x^4 - 8x^2 + 1$$

$w(x) = (1 - x^2)^{-\frac{1}{2}}$ funcția pondere

$$\int_{-1}^{1} w(x) T_m(x) T_n(x)\,dx = \int_{-1}^{1} \frac{\cos(m \cdot \arccos x) \cdot \cos(n \cdot \arccos x)}{\sqrt{1-x^2}} =$$

considerăm $\theta = \arccos x \Rightarrow d\theta = \frac{1}{\sqrt{1-x^2}}$

$$= \int_0^{\pi} \cos(n\theta)\cos(m\theta)\,d\theta \overset{m \neq n}{=\!=} \int_0^{\pi} \frac{1}{2}\left[\cos(m+n)\theta + \cos(m-n)\theta\right] d\theta =$$

$$= \frac{1}{2} \frac{\sin(m+n)\theta}{m+n}\Big|_0^{\pi} + \frac{1}{2} \frac{\sin(m-n)\theta}{m-n}\Big|_0^{\pi} = 0$$

$$\int_{-1}^{1} w(x) T_m^2(x)\,dx = \int_0^{\pi} \cos^2(m\theta)\,d\theta = \int_0^{\pi} \frac{1+\cos(2m\theta)}{2}\,d\theta = \frac{\theta}{2}\Big|_0^{\pi} + \frac{\sin(2\theta m)}{4}\Big|_0^{\pi} =$$

$$= \frac{\pi}{2}$$

$$\int_{-1}^{1} w(x) T_m(x) T_n(x)\,dx = \begin{cases} 0, & m \neq n \\ \frac{\pi}{2}, & m = n \end{cases}$$